Solar Na appendix

Sun · Na · downloadable appendix

Document	Solar Na appendix
Subject	Solar photospheric abundance appendix · Na
Author	Ryan Schmitt
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Web	https://exoplanetcodex.org/systems/sol/elements/na/

Solar sodium: 6.312 ± 0.039

Asplund, Amarsi & Grevesse 2021: 6.22 ± 0.03 · this measurement +0.092 dex

The three-dimensional non-LTE products apply published line-by-line corrections (Asplund et al. 2021; Canocchi et al. 2026) to our one-dimensional fits; they are not our own three-dimensional calculation. The H- and K-band products sit about 0.4 dex below the visible ones. They are shown as measured, with their full error bars.

Measured the way Asplund, Amarsi & Grevesse (2021) measure it: the Asplund et al. 2021 line set (the five Na I lines of their Table A.1), 5 lines measured on one spectrum, each part weighted by its uncertainty.

Na I: 6.291 ± 0.059 · Kitt Peak (1984, molecfit) · VIS · Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

Na I: 6.328 ± 0.052 · Kitt Peak (1984, molecfit) · NIR · Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

Highlighted Products

near-UV

6.671 ± 0.163

Na I · Deep-graded lines · 1D NLTE (Gerber) · Kitt Peak (Kurucz 2005) · Deep Grade

VIS

6.291 ± 0.059

Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

red-optical

6.338 ± 0.312

Na I · Codex-graded lines · 1D NLTE (Gerber) · Kitt Peak (1984, molecfit) · Codex Grade

NIR

6.328 ± 0.052

Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

H

5.779 ± 0.882

Na I · Codex-graded lines · 1D NLTE (Amarsi) · CRIRES+ H · Codex Grade

K

5.903 ± 0.182

Na I · Codex-graded lines · 1D NLTE (Amarsi) · CRIRES+ · Codex Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

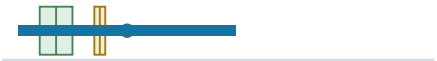
Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	30 products -- Na I · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	54 products -- Na I · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	28 products -- Na I · Deep-graded lines

Error-bar Forest Plot

near-UV — Kitt Peak (Kurucz 2005)
A(Na) dex; axis 6.12–6.91

Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.689 ±0.178
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 2 lines		6.574 ±0.186
Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.671 ±0.163
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.713 ±0.170

near-UV — Kitt Peak (1984, molecfit)
A(Na) dex; axis 6.12–6.91

Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.465 ±0.191
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 2 lines		6.350 ±0.199
Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.437 ±0.186
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.489 ±0.180

near-UV axis: 6.12 · 6.32 · 6.52 · 6.71 · 6.91 A(Na) dex

VIS — HARPS
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		6.278 ±0.113
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 4 lines		6.264 ±0.119
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 4 lines		6.226 ±0.120
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 4 lines		6.259 ±0.114
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 4 lines		6.282 ±0.113
Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.228 ±0.292
Na I · Deep-graded lines · 3D NLTE (Amarsi) Deep Grade · 2 lines		6.239 ±0.309
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 1 line		6.102 ±0.176
Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.218 ±0.370
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.243 ±0.242
Na I · Codex-graded lines · 1D LTE Codex Grade · 22 lines		6.370 ±0.077
Na I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 6 lines		6.264 ±0.094
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 22 lines		6.295 ±0.085
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 20 lines		6.320 ±0.067
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 22 lines		6.380 ±0.099

VIS — IAG FTS
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		6.394 ±0.065
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 1 line		6.386 ±0.073
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		6.362 ±0.072

VIS — IAG FTS
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 1 line		6.391 ±0.066
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line		6.401 ±0.065
Na I · Codex-graded lines · 1D LTE Codex Grade · 10 lines		6.354 ±0.137
Na I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 1 line		6.386 ±0.073
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 10 lines		6.318 ±0.142
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 10 lines		6.347 ±0.129
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 10 lines		6.353 ±0.129

VIS — IAG FTS
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines		6.293 ±0.050
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		6.281 ±0.060
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 3 lines		6.246 ±0.056
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 3 lines		6.277 ±0.046
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 3 lines		6.298 ±0.048
Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.191 ±0.247
Na I · Deep-graded lines · 3D NLTE (Amarsi) Deep Grade · 2 lines		6.202 ±0.267
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 1 line		6.066 ±0.159
Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.139 ±0.387
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.207 ±0.197
Na I · Codex-graded lines · 1D LTE Codex Grade · 6 lines		6.288 ±0.054

VIS — IAG FTS
A(Na) dex; axis 5.72–6.66

Na I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 5 lines		6.281 ±0.069
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 6 lines		6.238 ±0.060
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 6 lines		6.272 ±0.052
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 6 lines		6.292 ±0.054

VIS — Kitt Peak (Kurucz 2005)
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		6.300 ±0.050
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 4 lines		6.287 ±0.060
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 4 lines		6.248 ±0.059
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 4 lines		6.284 ±0.036
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 4 lines		6.305 ±0.049
Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.250 ±0.324
Na I · Deep-graded lines · 3D NLTE (Amarsi) Deep Grade · 2 lines		6.260 ±0.340
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 1 line		6.104 ±0.182
Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.236 ±0.395
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.263 ±0.259
Na I · Codex-graded lines · 1D LTE Codex Grade · 18 lines		6.300 ±0.078
Na I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 6 lines		6.287 ±0.063
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 18 lines		6.248 ±0.085
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 18 lines		6.284 ±0.078

VIS — Kitt Peak (Kurucz 2005)
A(Na) dex; axis 5.72–6.66

Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 18 lines		6.305 ±0.076

VIS — Kitt Peak (1984, molecfit)
A(Na) dex; axis 5.72–6.66

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		6.304 ±0.048
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 4 lines		6.291 ±0.059
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 4 lines		6.252 ±0.058
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 4 lines		6.278 ±0.051
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 4 lines		6.309 ±0.048
Na I · Deep-graded lines · 1D LTE Deep Grade · 2 lines		6.221 ±0.271
Na I · Deep-graded lines · 3D NLTE (Amarsi) Deep Grade · 2 lines		6.232 ±0.289
Na I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 1 line		6.095 ±0.178
Na I · Deep-graded lines · 1D NLTE (Gerber) Deep Grade · 2 lines		6.210 ±0.320
Na I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 2 lines		6.235 ±0.211
Na I · Codex-graded lines · 1D LTE Codex Grade · 17 lines		6.294 ±0.083
Na I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 6 lines		6.291 ±0.062
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 17 lines		6.247 ±0.089
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 16 lines		6.263 ±0.066
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 17 lines		6.298 ±0.077

VIS axis: 5.72 · 5.96 · 6.19 · 6.43 · 6.66 A(Na) dex

red-optical — IAG FTS
A(Na) dex; axis 4.06–7.92

Na I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		6.327 ±0.452
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		5.990 ±1.902
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 4 lines		6.332 ±0.449

red-optical — Kitt Peak (Kurucz 2005)
A(Na) dex; axis 4.06–7.92

Na I · Codex-graded lines · 1D LTE Codex Grade · 4 lines		6.322 ±0.381
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.231 ±0.700
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 4 lines		6.271 ±0.417
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 4 lines		6.327 ±0.376

red-optical — Kitt Peak (1984, molecfit)
A(Na) dex; axis 4.06–7.92

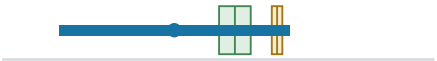
Na I · Codex-graded lines · 1D LTE Codex Grade · 5 lines		6.364 ±0.312
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines		6.262 ±0.624
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 5 lines		6.338 ±0.312
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 5 lines		6.370 ±0.300

red-optical axis: 4.06 · 5.02 · 5.99 · 6.96 · 7.92 A(Na) dex

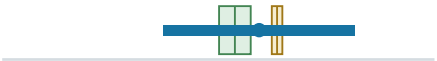
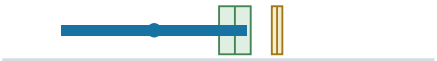
NIR — CRIRES+ J
A(Na) dex; axis 5.78–6.60

Na I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.213 ±0.355
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 1 line		5.901 ±0.092
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		6.196 ±0.359
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		6.215 ±0.349

NIR — CRIRES+ Y (full arm)
A(Na) dex; axis 5.78–6.60

Na I · Codex-graded lines · 1D LTE Codex Grade · 1 line		6.123 ±0.217
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 1 line		6.105 ±0.219
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 1 line		6.108 ±0.227
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 1 line		6.122 ±0.217

NIR — Kitt Peak (1984, molecfit)
A(Na) dex; axis 5.78–6.60

Na I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line		6.301 ±0.039
Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 1 line		6.328 ±0.052
Na I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line		6.295 ±0.049
Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 1 line		6.266 ±0.182
Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line		6.303 ±0.038
Na I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		6.252 ±0.163
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.067 ±0.176
Na I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 3 lines		6.232 ±0.161
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		6.251 ±0.160

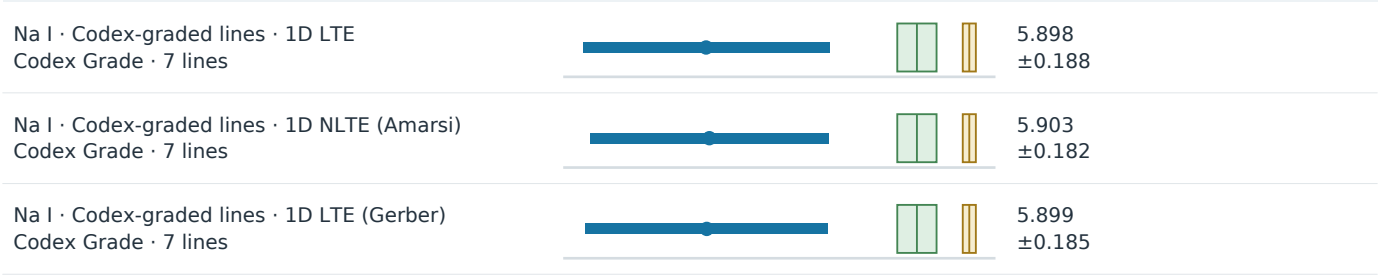
NIR axis: 5.78 · 5.98 · 6.19 · 6.39 · 6.60 A(Na) dex

H — CRIRES+ H
A(Na) dex; axis 4.85–6.76

Na I · Codex-graded lines · 1D LTE Codex Grade · 3 lines		5.796 ±0.881
Na I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 3 lines		5.779 ±0.882
Na I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 3 lines		5.803 ±0.927

H axis: 4.85 · 5.32 · 5.80 · 6.28 · 6.76 A(Na) dex

K — CRIRES+
A(Na) dex; axis 5.68–6.34



K axis: 5.68 · 5.84 · 6.01 · 6.17 · 6.34 A(Na) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the total uncertainty -- the statistical and systematic terms combined in quadrature, as Asplund et al. 2021 and Lodders et al. 2025 quote it. The full error budget remains attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every sodium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the sodium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Na)	Largest uncertainty terms (dex)
near-UV	Na I · Deep-graded lines	Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2	6.350–6.671 (1D-NLTE)	fit / line-to-line scatter 0.14, microturbulence 0.08
VIS	Na I · Deep-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	1–2	6.202–6.260 (3D-NLTE)	fit window 0.25, model atmosphere 0.08
VIS	Na I · Codex-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	1–22	6.264–6.386 (3D-NLTE)	continuum placement 0.03, fit / line-to-line scatter 0.03
VIS	Na I · Asplund et al. 2021 lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	1–4	6.264–6.386 (3D-NLTE)	continuum placement 0.04, NLTE / 3D correction 0.03

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Na)	Largest uncertainty terms (dex)
red-optical	Na I · Codex-graded lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	2–5	5.990–6.338 (1D-NLTE)	continuum placement 0.22, fit window 0.20
NIR	Na I · Codex-graded lines	CRIRES+ J, CRIRES+ Y (full arm), Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE	1–3	5.901–6.232 (1D-NLTE)	fit / line-to-line scatter 0.14, fit window 0.07
NIR	Na I · Asplund et al. 2021 lines	Kitt Peak (1984, molecfit)	1D-LTE, 1D-NLTE, 3D-NLTE	1	6.328 (3D-NLTE)	NLTE / 3D correction 0.03, fit / line-to-line scatter 0.02
H	Na I · Codex-graded lines	CRIRES+ H	1D-LTE, 1D-NLTE	3	5.779 (1D-NLTE)	fit / line-to-line scatter 0.69, telluric residual 0.54
K	Na I · Codex-graded lines	CRIRES+	1D-LTE, 1D-NLTE	7	5.903 (1D-NLTE)	fit / line-to-line scatter 0.13, telluric residual 0.12

Solar sodium context

Sodium is traced by neutral Na I lines from the near-ultraviolet to the K band. The reference lines are the five Asplund et al. 2021 used (4751, 5148, 6154, 6160 and 10746 Å), with the oscillator strengths they adopted. Na I departs from LTE, so each spectrum also carries non-LTE products; each set, spectrum and treatment remains a separate product.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Na appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit e1e0790a0580431c163a1dc7cef0c14f86d038da
Feed Na.json v1.131 · updated 2026-10-27T15:01:19Z
Generated 2026-10-28T15:47:36-06:00 (source commit timestamp)
Feed SHA-256 0b09aec1b38fe26d0250d2e8bce185e8703d8398ceb3a180d70d87d33356a1ba
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Na comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. Na I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [4].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Na_Amarsi2020_PySME.csv). Sources: [5].

Model and grid sources. Feed treatment: synth-1D-LTE-gerber; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

Model and grid sources. Feed treatment: ENGINE-B-NLTE; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

Solar abundance and indicator analyses. Published line set NA_AGSS21 (the line selection). Sources: [1].

Model and grid sources. Cited solar abundance differences from the published model columns, applied additively to our LTE fits; not a new in-transfer 3D calculation. Baseline atmosphere: A_marcs_lte, A_1d_lte Sources: [1], [6].

Model and grid sources. Cited solar abundance differences from the published model columns, applied additively to our LTE fits; not a new in-transfer 3D calculation. Baseline atmosphere: A_marcs_lte Sources: [1].

Product-to-source index

H · CRIRES+ H · Na I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Na-products-fb210d1eee02

H · CRIRES+ H · Na I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Na-products-a80b76db5d6a

H · CRIRES+ H · Na I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Na-products-c047829fd0f2

K · CRIRES+ · Na I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Na-products-0552befecb14

K · CRIRES+ · Na I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Na-products-4f1dee816393

K · CRIRES+ · Na I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Na-products-caab8cbc99e0

NIR · CRIRES+ J · Na I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Na-products-1229a0a156f8

NIR · CRIRES+ J · Na I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Na-products-f1c4ef0ec48e

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NIR · CRIRES+ Y (full arm) · Na I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Na-products-85aee27b99bb

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NIR · Kitt Peak (1984, molecfit) · Na I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Na-products-bd12d76eaf22

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VIS · HARPS · Na I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Na-products-65aef47bc520

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VIS · HARPS · Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Na-products-9439cb57fb7e

VIS · HARPS · Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [3], [4] Na-products-5919b5d71e97

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VIS · HARPS · Na I · Deep-graded lines · 3D NLTE (Amarsi) · [1], [2], [3], [6] Na-products-cd4f141ff187

VIS · HARPS · Na I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Na-products-77cd03c01180

VIS · HARPS · Na I · Deep-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Na-products-252fafda80c1

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VIS · HARPS · Na I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Na-products-4c88219f1ea2

VIS · HARPS · Na I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [3], [6] Na-products-034d202da814

VIS · HARPS · Na I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3] Na-products-1a483ec17b8d

VIS · HARPS · Na I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Na-products-a3583bba6864

VIS · HARPS · Na I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Na-products-abce6000c004

VIS · IAG FTS · Na I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [4] Na-products-541e815eee69

VIS · IAG FTS · Na I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [2], [6] Na-products-e2653a1dbfc3

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VIS · IAG FTS · Na I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [4] Na-products-ecad184cc0b3

VIS · IAG FTS · Na I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [4] Na-products-80ede85d09ae

VIS · IAG FTS · Na I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Na-products-ce9ac5f39f49

VIS · IAG FTS · Na I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [2], [6] Na-products-ab09b0d70813

VIS · IAG FTS · Na I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3] Na-products-348c83d33858

VIS · IAG FTS · Na I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Na-products-e50beeefb79a

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VIS · IAG FTS · Na I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Na-products-bf07027dc13e

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VIS · IAG FTS · Na I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3] Na-products-28f9f215407b

VIS · IAG FTS · Na I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Na-products-a35858a340fb

VIS · IAG FTS · Na I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Na-products-ad0c84319fcb

VIS · Kitt Peak (Kurucz 2005) · Na I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Na-products-4569604fa768

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red-optical · Kitt Peak (1984, molecfit) · Na I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Na-products-f78f5834d3bb

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red-optical · Kitt Peak (1984, molecfit) · Na I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Na-products-1582fe7c0f11

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

[2] — Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites. Space Science Reviews 221 DOI: 10.1007/s11214-025-01146-w. source record

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. source record

Model and grid sources

[4] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

[5] — Amarsi, A. M.; Lind, K.; Osorio, Y.; Nordlander, T.; Bergemann, M.; et al. (2020). The GALAH Survey: non-LTE departure coefficients for large spectroscopic surveys. A&A 642, A62 DOI: 10.1051/0004-6361/202038650. source record

[6] — Canocchi, G.; Wang, E. X.; Amarsi, A. M.; Lind, K.; Racca, M. (2026). 3D NLTE Sodium abundances in late-type stars. arXiv 2603.22975 DOI: 10.5281/zenodo.19201830. source record